Exercises

Submit your answers here: *https://forms.gle/2ToaAAthV7qX9Gxs9*

Exercise 1

Consider a polynomial function $p(x)$ such that it passes exactly through the following five points:

$$(-1,-5), \quad (0,1), \quad (1,1), \quad (2,7), \quad (3,31).$$

Determine explicitly the polynomial $p(x)$ that satisfies these conditions and has minimal possible degree.

Exercise 2

Consider the nonlinear system:

$$\begin{cases} x^2 + y^2 - 8 = 0, \\ e^x + y - e - 1 = 0. \end{cases}$$

Compute an approximate numerical solution (one is enough, even if the system has more than one solution).

Exercise 3

You are given an image (grayscale, only one color-dimension). Construct a Python algorithm that generates a new image which represents a rank-reduced approximation of the original image, using only 10% of the numerical rank of the matrix corresponding to the original image.

https://polybox.ethz.ch/index.php/s/fjdASWRcebran2F